

TrES-1b

R-1902

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_150707 · created 2026-08-05T22:59:07+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457210.7377 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.053
Transit depth (Rp/Rs)^2	3.1 +/- 1.86 [%]
Orbital Inclination (inc)	85.91 +/- 1.44
Airmass coefficient 1 (a1)	4.48 +/- 3.0
Airmass coefficient 2 (a2)	-0.99 +/- 0.98
Scatter in the residuals of the lightcurve fit is	65.6 %
Best Comparison Star	#2 - [539, 355]
Optimal Aperture	0.83
Optimal Annulus	7.16
Transit Duration (day)	0.077 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.053 vs 0.1358 (deviation 0.76σ)
Duration fit vs published	0.077 d vs 0.1042 d (deviation 1.01σ)
Mid-time O–C	-3.5 min
Depth SNR	3.3
Residual scatter	65.6 %

Light curve

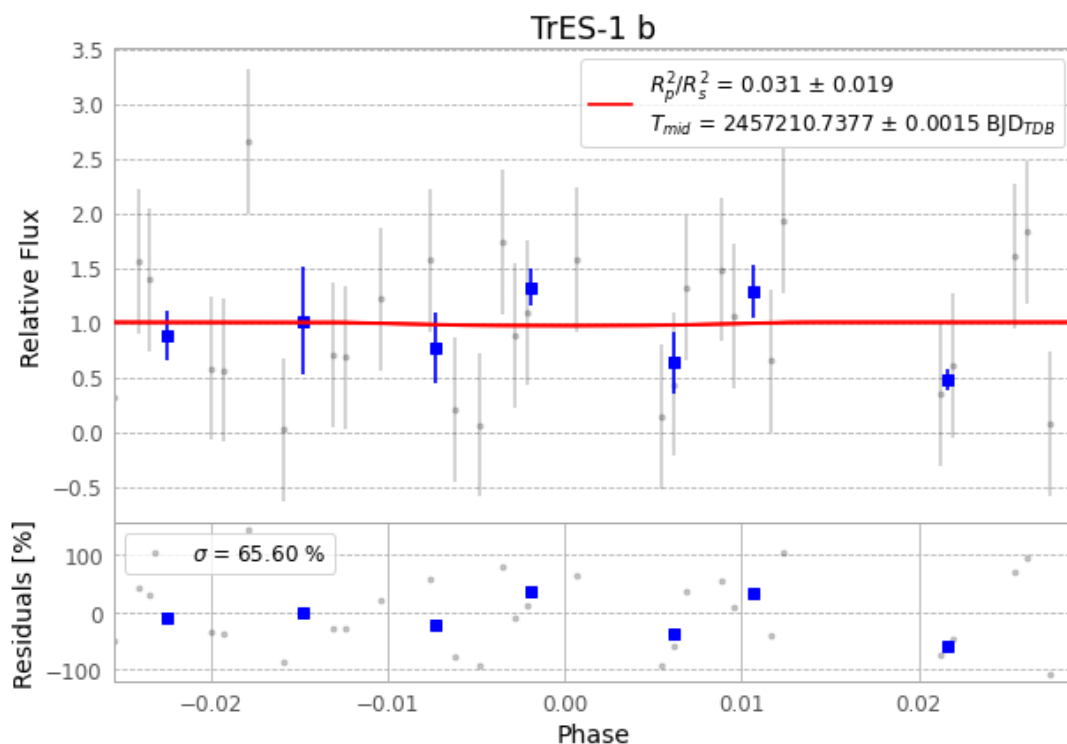


Figure 1 — FinalLightCurve_TrES-1 b_2015-07-06.png (SHA-256 8070d1974e69287d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [474, 428], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 722972eb3080d19d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), TRES-1150707034212.FITS → TRES-1150707080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-150707.log (f12f96a8f715...), FinalLightCurve_TrES-1 b_2015-07-06.png (8070d1974e69...), FinalLightCurve_TrES-1 b_2015-07-06.csv (70b953901373...)

Compute resources

Wall time 178.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 22:59Z.